

Ray tracing – Curved quadratic elements

1. Intersection problem

Surface position on the tetrahedron's face (weighted sum of shape functions):

$$P(g, h) = \sum_{i=0}^5 N_i(g, h) \cdot f_{nodes_i} \quad (1)$$

Ray:

$$R(t) = \mathbf{o} + t\mathbf{d} \quad (2)$$

where $\mathbf{d}$ is the ray's direction, and $\mathbf{o}$ is the ray's origin

Face-ray intersection (3 unknowns):

$$\mathbf{o} + t\mathbf{d} - P(g, h) = 0 \quad (3)$$

Nonlinear system of equations:

$$F(t, g, h) = \mathbf{o} + t\mathbf{d} - \sum_{i=0}^5 N_i(g, h) \cdot f_{nodes_i} = \mathbf{0} \quad (4)$$

2. Möller-Trumbore algorithm for linear triangle intersection (initial guess)

$$\mathbf{E}_1 = V_1 - V_0, \mathbf{E}_2 = V_2 - V_0 \quad (5)$$

$$\mathbf{p} = \mathbf{d} \times \mathbf{E}_2 \quad (6)$$

$$det = \mathbf{E}_1 \cdot \mathbf{p} \quad (7)$$

$$t_{vec} = \mathbf{o} - V_0 \quad (8)$$

Where V_0, V_1, V_2 are triangle vertices.

$$u = \frac{\mathbf{t}_{vec} \cdot \mathbf{p}}{det} \quad (9)$$

$$\mathbf{q} = \mathbf{t}_{vec} \times \mathbf{E}_1 \quad (10)$$

$$v = \frac{\mathbf{d} \cdot \mathbf{q}}{det} \quad (11)$$

$$t = \frac{\mathbf{E}_2 \cdot \mathbf{q}}{det} \quad (12)$$

3. Newton-Raphson

Jacobian of F :

$$\mathbf{J}_F = \left[\frac{\partial F}{\partial f}, \frac{\partial F}{\partial g}, \frac{\partial F}{\partial h} \right] \quad (13)$$

$$\mathbf{J}_F = \left[\mathbf{d}, -\frac{\partial P}{\partial g}, -\frac{\partial P}{\partial h} \right] \quad (14)$$

$$\frac{\partial P}{\partial g} = \sum_{i=0}^5 \frac{\partial N_i}{\partial g} \cdot f_{nodes_i} \quad (15)$$

$$\frac{\partial P}{\partial h} = \sum_{i=0}^5 \frac{\partial N_i}{\partial h} \cdot f_{nodes_i} \quad (16)$$

The update by solving a linear system:

$$\mathbf{\Delta} = [\Delta t, \Delta g, \Delta h]^T \quad (17)$$

$$\mathbf{J}_F \cdot \mathbf{\Delta} = -F(t, g, h) \quad (18)$$

$$\begin{bmatrix} \mathbf{d}_x & -\frac{\partial P_x}{\partial g} & -\frac{\partial P_x}{\partial h} \\ \mathbf{d}_y & -\frac{\partial P_y}{\partial g} & -\frac{\partial P_y}{\partial h} \\ \mathbf{d}_z & -\frac{\partial P_z}{\partial g} & -\frac{\partial P_z}{\partial h} \end{bmatrix} \begin{bmatrix} \Delta t \\ \Delta g \\ \Delta h \end{bmatrix} = -(\mathbf{o} + t\mathbf{d} - P(g, h)) \quad (19)$$

4. Surface normal

$$\boldsymbol{n} = \frac{\partial P}{\partial g} \times \frac{\partial P}{\partial h} \tag{20}$$